

QUARK SVD first results — $m_s/m_d = \text{rank}^2 \cdot n_C = 20 \rightarrow$
 DERIVED (the first quark mass ratio at Derived, promotes from
 Conditional; forced, target-innocent, clears the blind bar);
 anchor $3\pi/128$ exact; top-ceiling Derived. FLAGSHIP clears
 Keeper's final glance — ship-ready pending Cal's 60-sec abstract
 re-verify.

Keeper

2026-07-29

K993 — The quark sector's first Derived, and the flagship clears

★★ $m_s/m_d = \text{rank}^2 \cdot n_C = 20 \rightarrow$ DERIVED (the first quark mass ratio at Derived)

Elie's $\mathbb{C}^3$ SVD (toy 4914) on Lyra's pinned addresses (angular $(\frac{1}{2}, \frac{1}{2})$ shared; radial $n=0,1,2$ = the three Korányi-Wolfstrata) produced, target-innocently: - anchor $\|f_0\|^2 = \Gamma(5/2)^2/\Gamma(5) = 3\pi/128$ — verified EXACT (Keeper cross-check). - $m_s/m_d = \text{rank}^2 \cdot n_C = 4 \cdot 5 = 20$ — the down-type is a direct one-step radial ratio ($n=0 \rightarrow 1$); its Gindikin integer content is forced by $\{\text{rank}=2, n_C=5\}$. Verified 0.00% vs observed ($93.4/4.67 = 20.00$). Both structural integers, forced from the domain; Elie did NOT calibrate to the mass — it fell out $\rightarrow$ clears the blind forward bar (the same one the tau's 71 failed). - RULING: $m_s/m_d \rightarrow$ DERIVED (promotes from CONDITIONAL — the old “overlap-vs-norm gate” is resolved: it's the radial-overlap ratio, computed from the pinned addresses). The first quark mass ratio at Derived, pending Cal's blind-bar confirmation (that it clears the same target-innocence the leptons did — it does: not calibrated, forced from $\{\text{rank}, n_C\}$). - This also validates the addresses are pinned right (the clean forced 20 is the check).

★ Top-ceiling $\rightarrow$ DERIVED (unchanged, re-anchored)

$m_t \leq v/\sqrt{2} = 174.1$ GeV (Cauchy-Schwarz on the rank-1 overlap); obs 172.69 sits just under ($y_t = 0.992$). The SVD structure falls out as it should: one leading singular value = the top (rank-1 condensate); everything else ($c, u, b, s, d + \text{CKM}$) = the off-rank-1 correction spectrum, Tier-2 by construction.

★ The rest of the quark hierarchy — HELD Tier-2 (pending Lyra's forced Pochhammer args)

Elie correctly refused to calibrate the Gindikin coefficients to the light-quark masses / mixing angles (the fit-in-disguise trap). The up-type is a ratio-of-ratios (Tier-2 continuous); the precise numbers wait on Lyra's forced Pochhammer arguments, not on tuning. Over-fits stay rejected ($m_c/m_u=588, m_t/m_c=137, m_b/m_s$). Any new Tier-1 clears the blind forward bar. On the colored sector, the restraint IS the credibility (Cal's watch: a colored ratio coming out as a clean lepton-style closed form is an over-fit prior until it survives the bar).

★ FLAGSHIP — clears Keeper's final glance; ship-ready

- Lyra applied all 5 §135 fixes (v0.2, grep-verified never-bare muon; 4 under-sells promoted to the ratified record; K992 electron detail folded).

- Grace caught a genuine ship-blocker: the muon/tau BASE tier fields in the ledger were stale (muon=IDENTIFIED, tau=FITTED — pre-banking), even though the k962 fields were correct. Fixed (muon→D, tau→I, qualifiers intact). Data layer now agrees with the paper — a hostile-reader inconsistency avoided. (Symmetry: Cal caught the paper over-showing in the abstract; Grace caught the ledger under-showing — same drift, both fixed.)
- Keeper gates all closed: tiers == record (electron D, muon D(e=n), tau home-D/value-I, count D); $6\pi^5$ re-audited (K992); Value-3 double-verified (K991); ledger ship-consistent.
- Remaining before GO: Cal's 60-second re-verify of the fixed abstract (confirm the muon line carries the qualifier) — cheap insurance on the one over-claim he'd caught. Then Casey GO (repo-internal, not pushed).

What's next (quark program)

Lyra pins the forced Pochhammer arguments → Elie computes the full Tier-2 set: reproduce V_{us} target-innocently (blind), then predict V_{cb} , m_c/m_u , and the neutrino Δm^2 we didn't use — the falsifiable payoff. Keeper rules each against the confined-bulk bar; V_{us} blind-reproduction is the next make-or-break (does the SVD's angular part give the Cabibbo angle without calibration?).

— K993, Keeper, 2026-07-29. $m_s/m_d=20$ DERIVED (first quark mass ratio; forced, target-innocent, clears blind bar); anchor $3\pi/128$ exact; top-ceiling Derived; rest Tier-2 pending Lyra's Pochhammer args. Flagship ship-ready (ledger blocker fixed) pending Cal's 60-sec abstract re-verify → Casey GO. See [[Keeper_K992_6pi5_electron_RE_AUDIT_target_innocent_CONFIRMED_6_is_C2_Casimir_pi5_is_Hua_volume_Cal_135_flag_07-29]], Elie toy 4914, Lyra F718.